

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

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Forename(s)

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Candidate signature

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INTERNATIONAL GCSE BIOLOGY

Paper 1

Thursday 7 November 2019

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against the answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



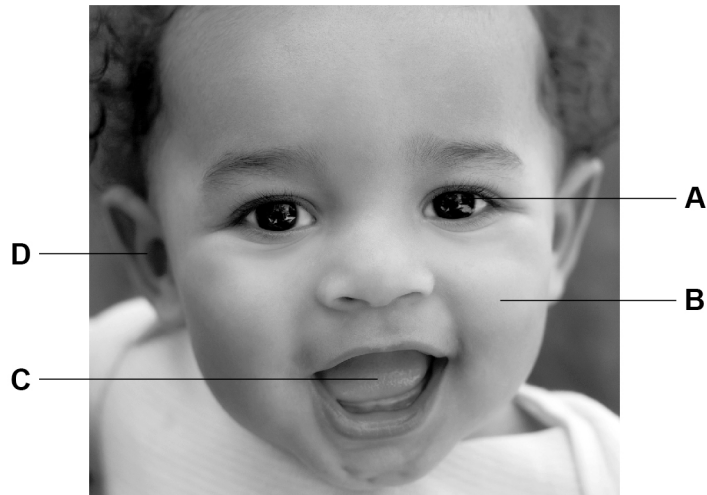
Answer **all** questions in the spaces provided.

0 1

This question is about the nervous system.

Figure 1 shows some sense organs of a baby.

Figure 1



Each sense organ has receptors that detect stimuli.

0 1 . 1

Draw **one** line from each stimulus to the sense organ which detects it.

[3 marks]

Stimulus	Sense organ
Chemical	A
Light	B
Sound	C
	D



0 1 . 2 Complete the sentence about reflex actions.

Choose answers from the box.

[2 marks]

automatic	conscious	painful	permanent	rapid
------------------	------------------	----------------	------------------	--------------

Reflex actions are _____ and _____.

0 1 . 3 Reflex actions are important for the survival of animals.

Give **one** example of a reflex action.

[1 mark]

0 1 . 4 Which type of neurone passes information from the receptor to the spinal cord in a reflex?

Tick (✓) **one** box.

[1 mark]

Motor neurone

☐

Relay neurone

☐

Sensory neurone

☐

0 1 . 5 Effectors carry out responses.

Name **one** type of effector.

[1 mark]

Question 1 continues on the next page

Turn over ►



0 1 . 6

Complete the sentence.

[1 mark]

Information is transferred across a synapse as _____.

0 1 . 7

Name the **two** parts of the central nervous system (CNS).

[1 mark]

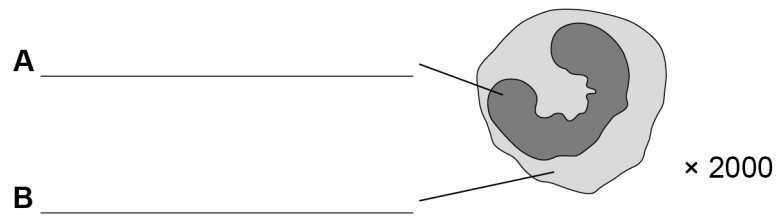
1 _____

2 _____

10



0 2

Figure 2 shows a white blood cell.**Figure 2**

0 2

. 1

Label parts **A** and **B** on **Figure 2**.**[2 marks]**

0 2

. 2

In **Figure 2** the image width of the white blood cell is 25 mm.

The white blood cell has been magnified 2000 times.

Calculate the actual width of the white blood cell.

Use the equation:

$$\text{Actual width} = \frac{\text{image width}}{\text{magnification}}$$

[2 marks]

Actual width = _____ mm

Question 2 continues on the next page**Turn over ►**

0 2 . 3 White blood cells are part of the human body's defence system against pathogens.

One type of pathogen is fungi.

Name **two** other types of pathogen.

[2 marks]

1 _____

2 _____

Vaccines are used to immunise a person against pathogens.

The MMR vaccine is used to immunise children against measles.

0 2 . 4 Name **one** other disease MMR immunises against.

[1 mark]

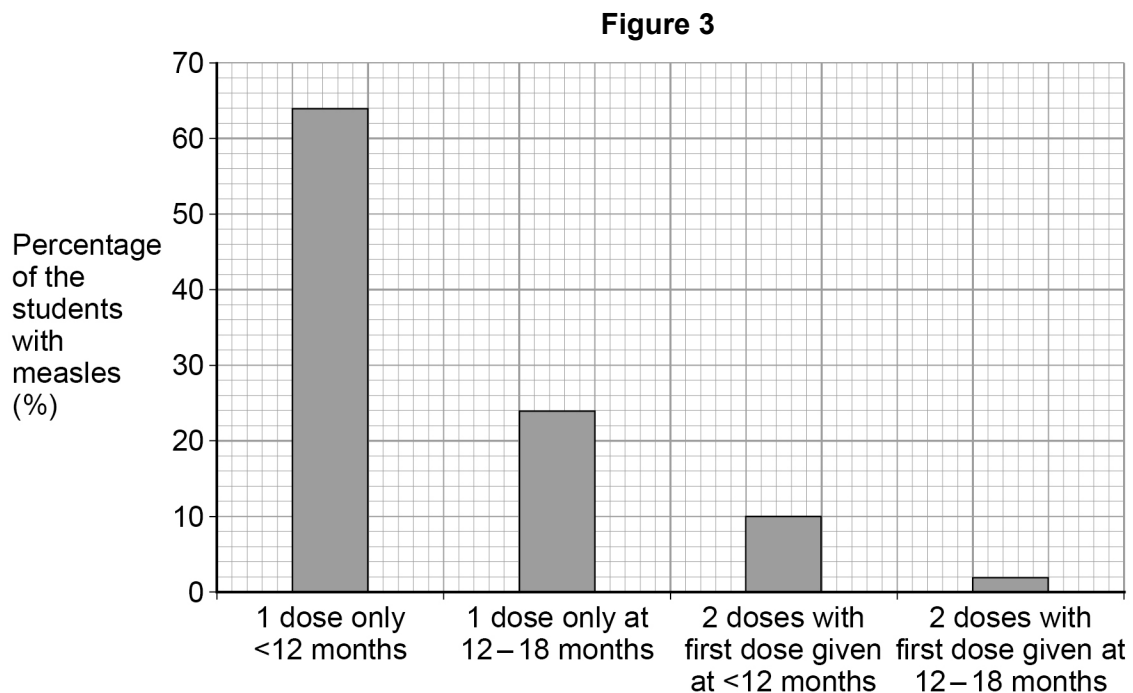
In the early 1990s a number of students at the same school developed measles.

All the students had received the MMR vaccine.

The MMR vaccine should be given in 2 doses:

- at 12–18 months
- at 5 years.

Figure 3 shows at what age each student with measles had received doses of the MMR vaccine.



0 2 . 5

How does **Figure 3** show that the MMR vaccine is more effective if 2 doses are given?

[1 mark]

0 2 . 6

Describe how **Figure 3** shows that the first dose of the MMR vaccine should be given at 12–18 months.

[2 marks]

10

Turn over for the next question

Turn over ►



0 3

Plants and animals both compete for resources.

0 3 . 1Name **two** resources that animals compete for.**[2 marks]**

1 _____

2 _____

0 3 . 2

Plants compete for space.

Name **two** other resources that plants compete for.**[2 marks]**

1 _____

2 _____

0 3 . 3

Competition between crop plants and weeds in a field results in lower yields of the crop plants.

Explain why the yield of crop plants is reduced by competition with weeds.

[3 marks]



Crop plants can be genetically engineered.

Genetic engineering involves removing a gene from one organism (**A**) and inserting the gene into another organism (**B**).

0 3 . 4 What is used to remove the gene from organism **A**?

Tick (✓) **one** box.

[1 mark]

Enzyme

☐

Hormone

☐

Virus

☐

0 3 . 5 Describe how the gene is inserted into organism **B**.

[2 marks]

0 3 . 6 A herbicide is a chemical that kills plants.

Crop plants can be genetically engineered to be resistant to herbicides.

Explain how using herbicide-resistant crop plants will increase the yield of the crop.

[2 marks]

Question 3 continues on the next page

Turn over ►



0	3	.	7
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Genetically engineered crops are called GM crops.

Some people have concerns about GM crops.

Give **two** concerns about GM crops.

[2 marks]

1 _____

2 _____

14



0 4

Blood circulates through arteries, veins and capillaries in the human body.

0 4 . 1

Complete the sentence.

[2 marks]

The walls of arteries are made up of _____ and
_____ fibres.

0 4 . 2

Suggest why it is important that the walls of arteries are thicker than the walls of veins.

[1 mark]

0 4 . 3

Capillaries are important in the movement of substances into and out of cells.

Give **one** way the walls of capillaries differ from the walls of arteries and of veins.**[1 mark]**

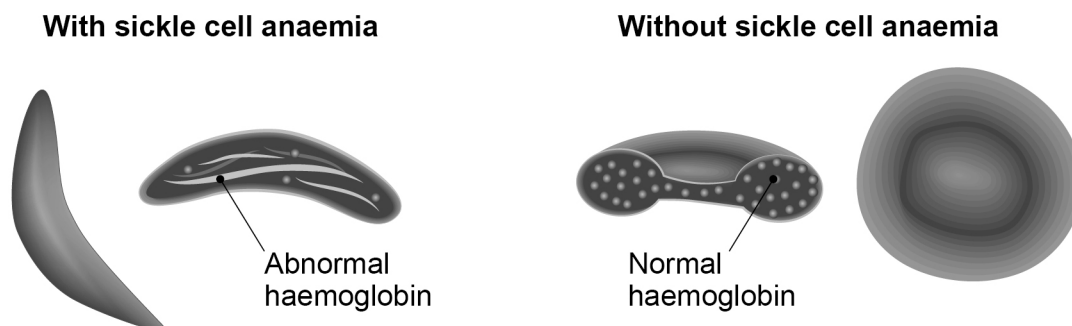
Question 4 continues on the next page**Turn over ►**

Some people have a disorder called sickle cell anaemia.

The haemoglobin inside red blood cells is abnormal in a person with sickle cell anaemia. The abnormal haemoglobin causes the cells to change shape and become stiff.

Figure 4 shows red blood cells from a person with sickle cell anaemia and from a person without sickle cell anaemia.

Figure 4



0 4 . 4

Suggest reasons why the tissues of a person with sickle cell anaemia may **not** get enough oxygen.

[3 marks]

0	4	.	5
---	---	---	---

Sickle cell anaemia is an inherited disorder caused by a recessive allele.

Two parents who do **not** have sickle cell anaemia have a child.

The child has sickle cell anaemia.

Draw a genetic diagram to show the probability that the parents' next child will also have sickle cell anaemia.

You should identify any of the children with sickle cell anaemia.

Use the following symbols:

A = allele for normal haemoglobin

a = allele for sickle cell haemoglobin

[4 marks]

Probability = _____

Question 4 continues on the next page

Turn over ►



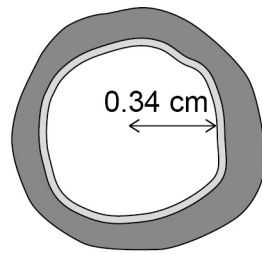
Some children who have sickle cell anaemia develop a second disease, **M**.

In disease **M** the lumen of the main artery supplying blood to the brain becomes very narrow.

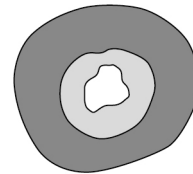
The lumen is the space in the middle of the artery.

Figure 5 shows two arteries from different people.

Figure 5



Healthy artery from a person
without disease **M**



Artery from a person
with disease **M**

The radius of the lumen of the healthy artery is 0.34 cm.

0 4 6

Calculate the cross-sectional area of the lumen of the healthy artery.

Use $\pi = 3.14$

Give your answer to 2 significant figures.

[3 marks]

Cross-sectional area = _____ cm^2



0 4 . 7

The rate of blood flow through an artery can be measured in cubic centimetres per second (cm^3/s).

It takes blood 0.29 seconds to travel a length of 2.0 cm through the artery of a person with disease **M**.

The cross-sectional area of the lumen of the artery is 0.012 cm^2

Calculate the rate of blood flow through the artery.

[3 marks]

Rate of blood flow = _____ cm^3/s

0 4 . 8

The rate of blood flow in an artery of a person with disease **M** is much lower than in a healthy artery.

This causes high blood pressure in the arteries in the brain.

Suggest why brain cells might die in a person with disease **M**.

[4 marks]

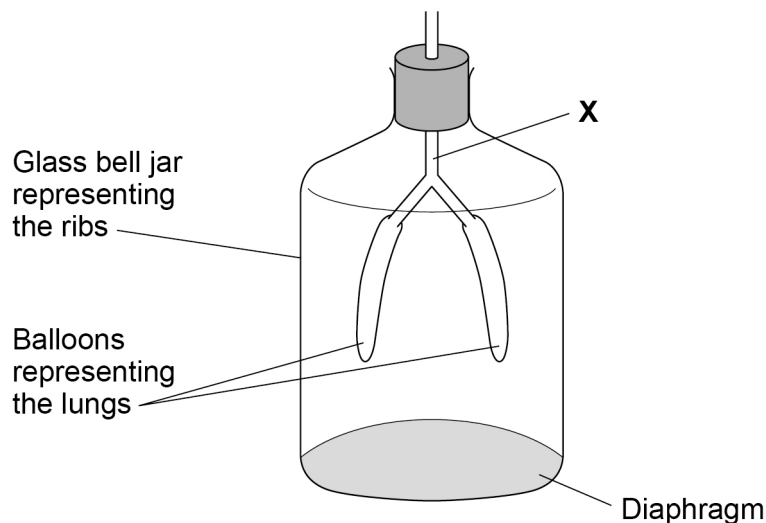


0 5

Figure 6 shows a model of the breathing system.

The different parts of the model represent different parts of the breathing system.

Figure 6



0 5 . 1

Which part of the breathing system does **X** represent?

[1 mark]

0 5 . 2

The model in **Figure 6** can be used to demonstrate breathing.

Complete the sentences.

[4 marks]

The rubber sheet being pulled down represents the diaphragm muscles _____.

This causes the volume of the glass jar to _____.

The pressure inside is now _____ than the pressure outside.

This causes air to _____ the balloons.



0 5 . 3

Tumours in the lung can be benign or malignant.

Give **two** ways benign tumours differ from malignant tumours.

[2 marks]

1 _____

2 _____

Tumours grow in the lungs of people with lung cancer.

CK-19 is a protein produced by tumours in lung cancer patients.

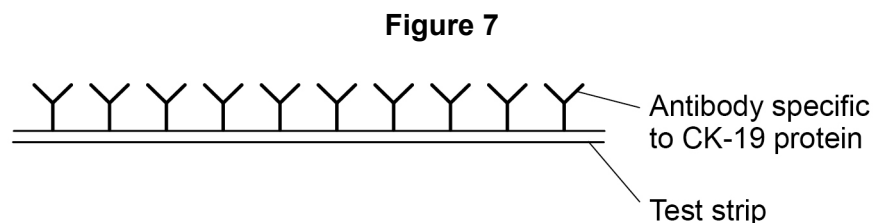
Scientists wanted to find out if CK-19 protein could be used as a tumour marker.
A tumour marker can show if a particular type of tumour is present.

Scientists investigated the levels of CK-19 protein in the blood of:

- 210 people with lung tumours
- 210 people without lung tumours.

The scientists used a test strip. The test strip has specific antibodies which detect the CK-19 protein.

Figure 7 shows a diagram of the test strip.



If the CK-19 protein is present in a person's blood it combines with the antibodies on the test strip to produce a colour.

The greater the intensity of the colour, the higher the concentration of CK-19 protein.

This is the method used.

1. Take a blood sample from a person.
2. Measure 0.7 cm^3 blood and add to the test strip.
3. Wait for one minute.
4. Wash the strip gently with water.
5. Measure the intensity of the colour on the strip.

Question 5 continues on the next page

Turn over ►

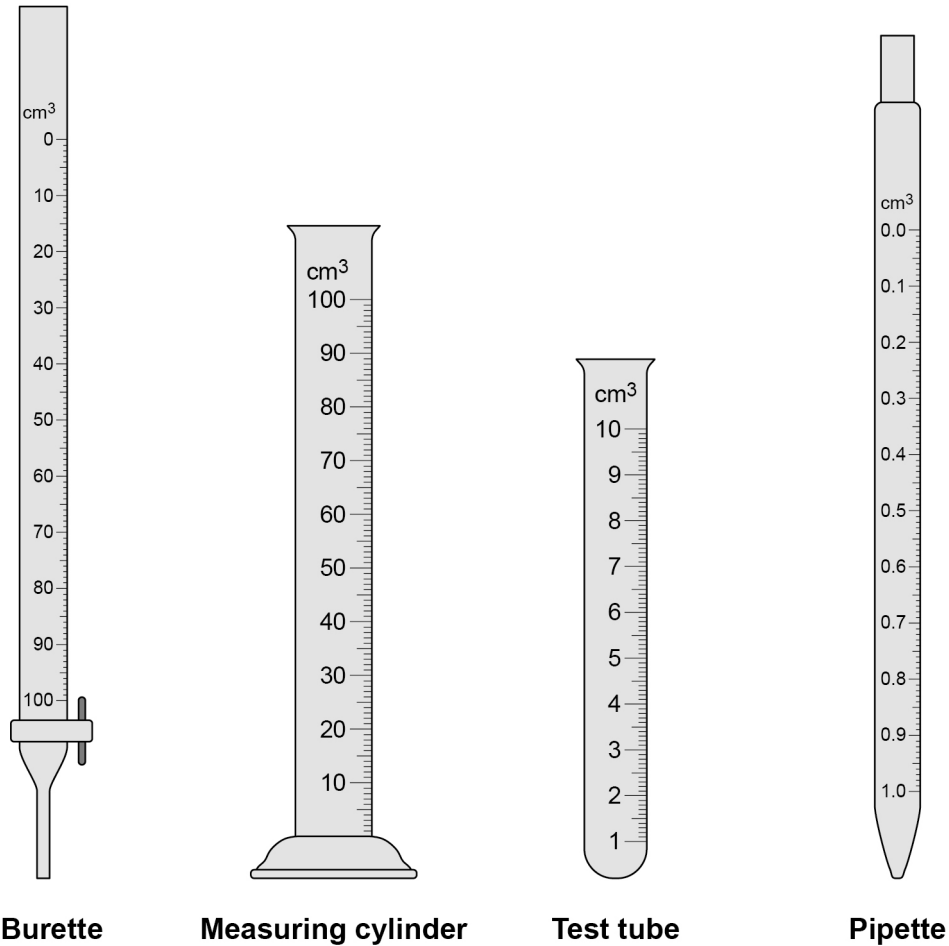


0 5 . 4

The scientists have different equipment available for measuring the 0.7 cm^3 blood sample.

Figure 8 shows 4 pieces of equipment that measure liquids.

Figure 8



Which piece of equipment should the scientists use to measure the blood sample?
Give a reason for your answer.

[2 marks]

Piece of equipment _____

Reason _____

0 5 . 5

Why was it important to test the levels of CK-19 protein in patients without tumours?

[1 mark]



0 5 . 6

The intensity of the colour did **not** continue to change after the test strip was washed.

Suggest why.

[1 mark]

0 5 . 7

Blood contains many different proteins.

Suggest why only the CK-19 protein binds to the antibody on the test strip.

[1 mark]

Question 5 continues on the next page

Turn over ►



The scientists divided the 420 patients into 6 equal groups.

- 3 groups with lung tumours
- 3 groups without lung tumours

Table 1 shows the scientists' results.

Table 1

Group	Mean concentration of CK-19 protein in mg / cm ³	
	With lung tumours	Without lung tumours
1	4.1×10^{-6}	1.6×10^{-6}
2	6.3×10^{-6}	2.9×10^{-6}
3	6.9×10^{-6}	2.0×10^{-6}
Mean for all patients		2.2×10^{-6}

0 5 . 8

Calculate the mean mass of CK-19 protein in a 0.7 cm^3 blood sample from patients **with** lung tumours.

[3 marks]

Mean mass of CK-19 protein = _____ mg



Evaluate the use of CK-19 protein as a tumour marker.

Use the data in Question 5.8 to help you.

[6 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

21

Turn over for the next question

Turn over ►



0	6
---	---

Kidneys are involved in homeostasis in the human body.

0	6	.	1
---	---	---	---

What is meant by homeostasis?

[1 mark]

0	6	.	2
---	---	---	---

Describe how a healthy kidney produces urine.

Do **not** refer to hormones in your answer.

[5 marks]



There is a link between kidney function and congestive heart failure (CHF).

When the ventricle walls do not contract normally CHF may result.

Cardiac output measures the total amount of blood the ventricles can pump out in 1 minute.

Cardiac output is calculated using the equation:

$$\text{Cardiac output} = \text{stroke volume} \times \text{heart rate}$$

Table 2 shows how heart rate and cardiac output differ in two groups of men.

Table 2

	Mean heart rate in beats / minute	Mean cardiac output in cm ³ / minute
60-year-old males with CHF	65	1300
60-year-old males without CHF	65	4550

0 6 . 3

Compare the stroke volume for 60-year-old males with CHF and 60-year-old males without CHF.

You should include calculations in your answer.

[2 marks]

Question 6 continues on the next page

Turn over ►



The hormone ADH is important for the functioning of the kidneys.

0 6 . 4 Which organ releases ADH?

Tick (✓) **one** box.

[1 mark]

Kidney

☐

Liver

☐

Pancreas

☐

Pituitary gland

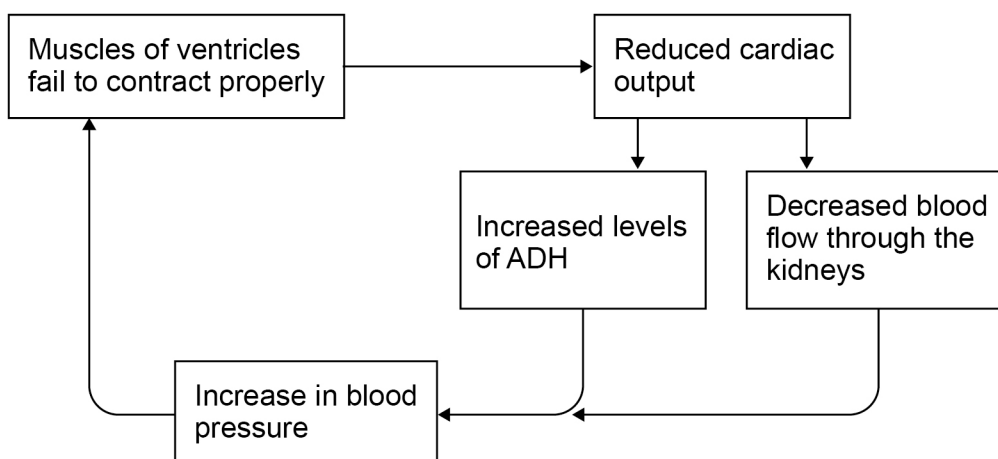
☐

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06.5 Some patients with CHF have increased levels of ADH in their blood.

Figure 9 shows some of the features of having CHF.

Figure 9



Explain how a reduced cardiac output causes the muscles of the ventricles to fail.

[5 marks]

[illegible]

14

END OF QUESTIONS



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